

Data Visualisation with ggplot2

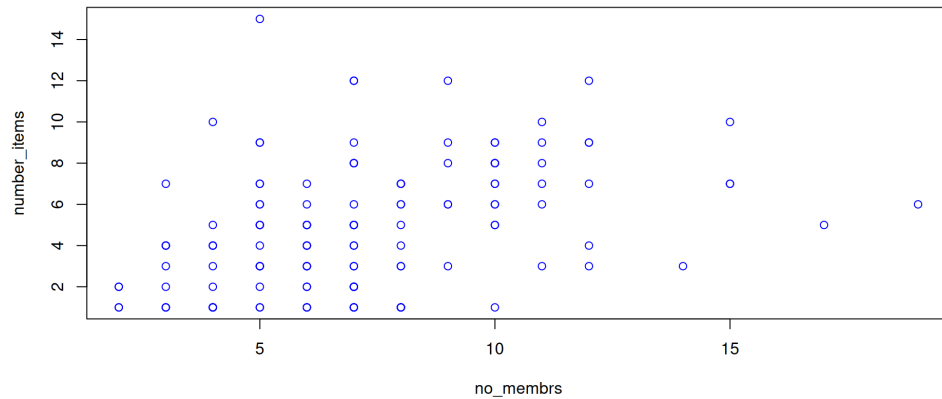
Data Carpentry for Social Sciences and Humanities

2025-10-09

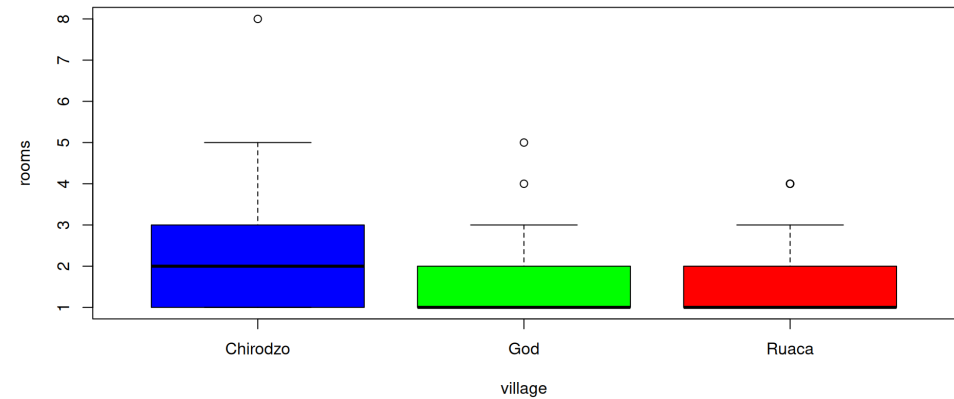
Why ggplot2?

...because these are 'base' plots

```
1 plot(number_items ~ no_membrs,  
2       interviews_plotting,  
3       col = "blue")
```



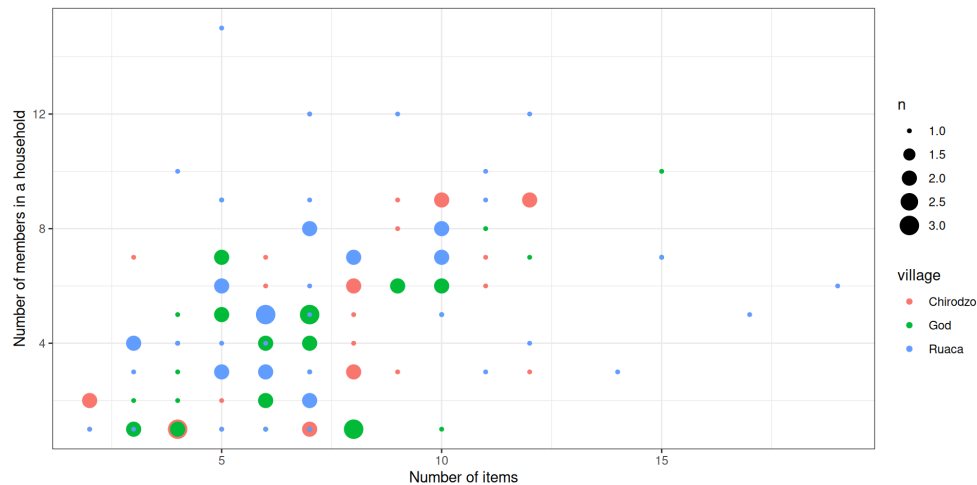
```
1 boxplot(rooms ~ village,  
2          interviews_plotting,  
3          col = c("blue", "green", "red"))
```



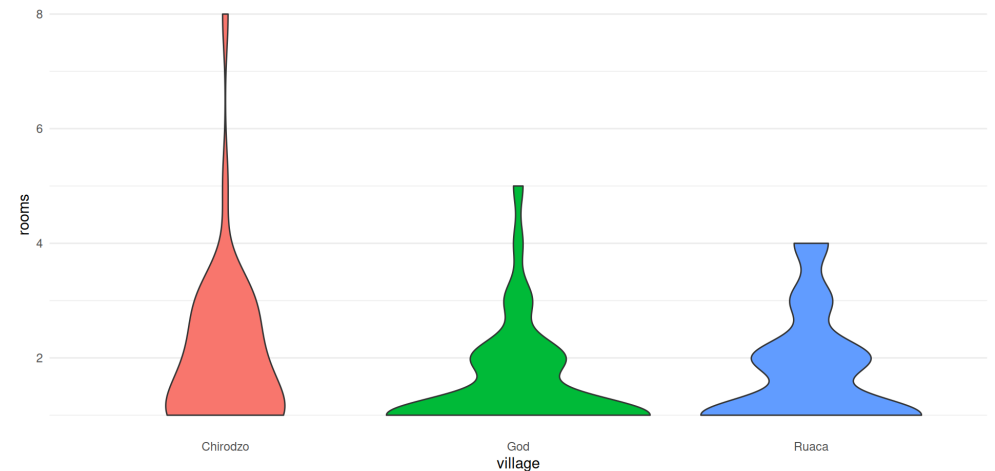
Why ggplot2?

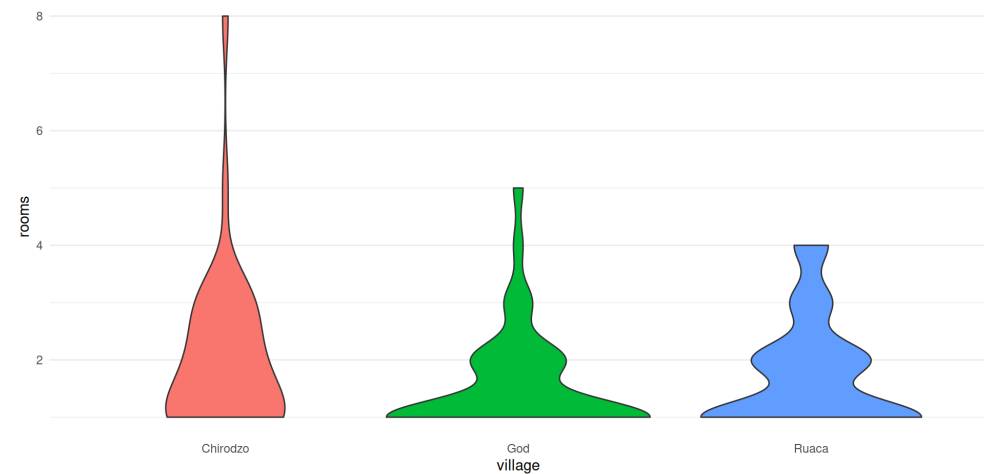
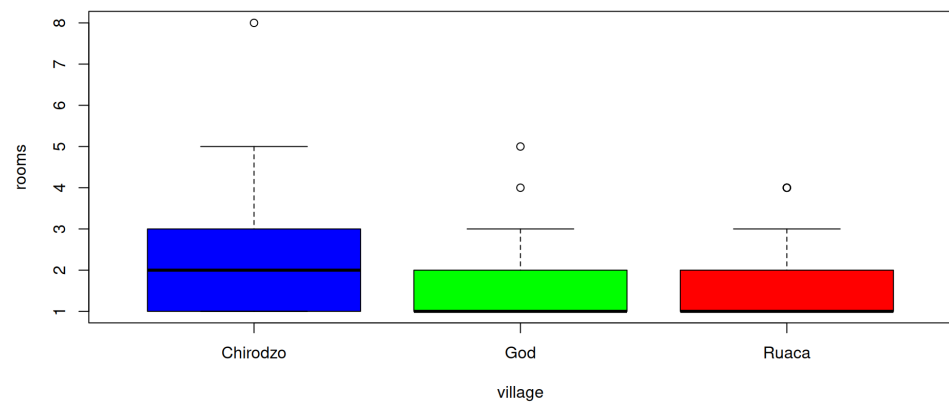
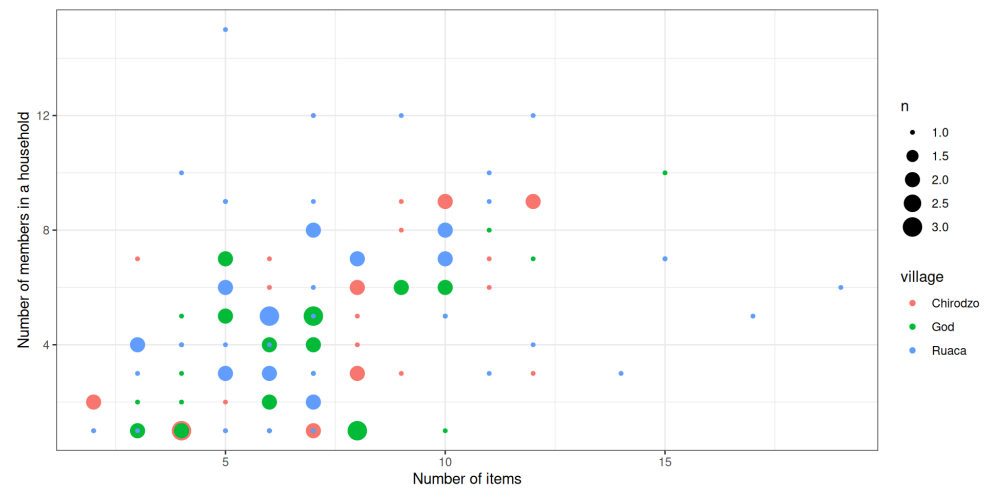
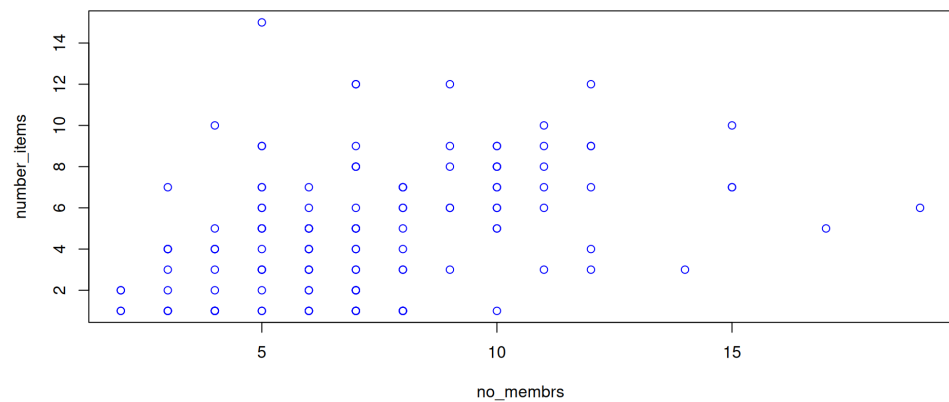
...and these are ggplots 😎

```
1 interviews_plotting %>%
2   ggplot(aes(x = no_membrs, y = number_item
3     geom_count() +
4     theme_bw() +
5     labs(x = "Number of items",
6       y = "Number of members in a house
```



```
1 interviews_plotting %>%
2   ggplot(aes(x = village, y = rooms, fill =
3     geom_violin() +
4     theme_minimal() +
5     theme(legend.position = "none",
6       panel.grid.major.x = element_blank()
```





ggplot2



ggplot2 is a package (included in **tidyverse**) for creating highly customisable plots that are built step-by-step by adding layers.

The separation of a plot into layers allows a high degree of flexibility with minimal effort.

```
<DATA> %>%  
  ggplot(aes(<MAPPINGS>)) +  
  <GEOM_FUNCTION>() +  
  <CUSTOMISATION>
```



Artwork by [@allison_horst](https://twitter.com/allison_horst)

Data Visualisation Exercises

Exercise 1

 6 mins

Create a new code chunk with the label `fig-rooms-scatter`.

Create a scatter plot of `rooms` by `village` with the `respondant_wall_type` showing in different colours.

Does this seem like a good way to display the relationship between these variables?

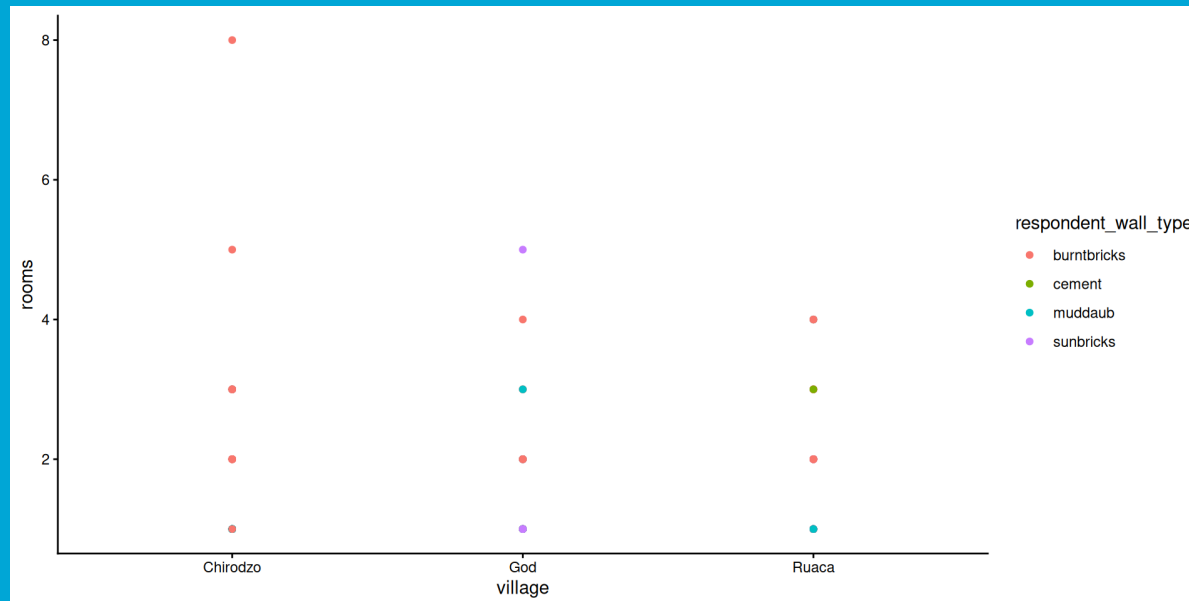
What other kinds of plots might you use to show this type of data?

06:00

Exercise 1: Solution

`geom_point` `geom_jitter`

```
1 interviews_plotting %>%  
2   ggplot(aes(x = village, y = rooms, colour = respondent_wall_type)) +  
3   geom_point() +  
4   theme_classic() +  
5   scale_fill_viridis_d() # add colourblind-friendly palette
```



Captioning

Now that we have created the plot, we can also create a caption using the `fig-cap` chunk option.

```
1 ```{r}
2 #| label: fig-rooms-scatter
3 #| fig-cap: "This plot shows the relationship between the variables room and village, but doesn't do a very good job at it."
4
5 interviews_plotting %>%
6   ggplot(aes(x = village, y = rooms, colour = respondent_wall_type)) +
7   geom_point() +
8   theme_classic() +
9   scale_fill_viridis_d() # add colourblind-friendly palette
10 ```
```

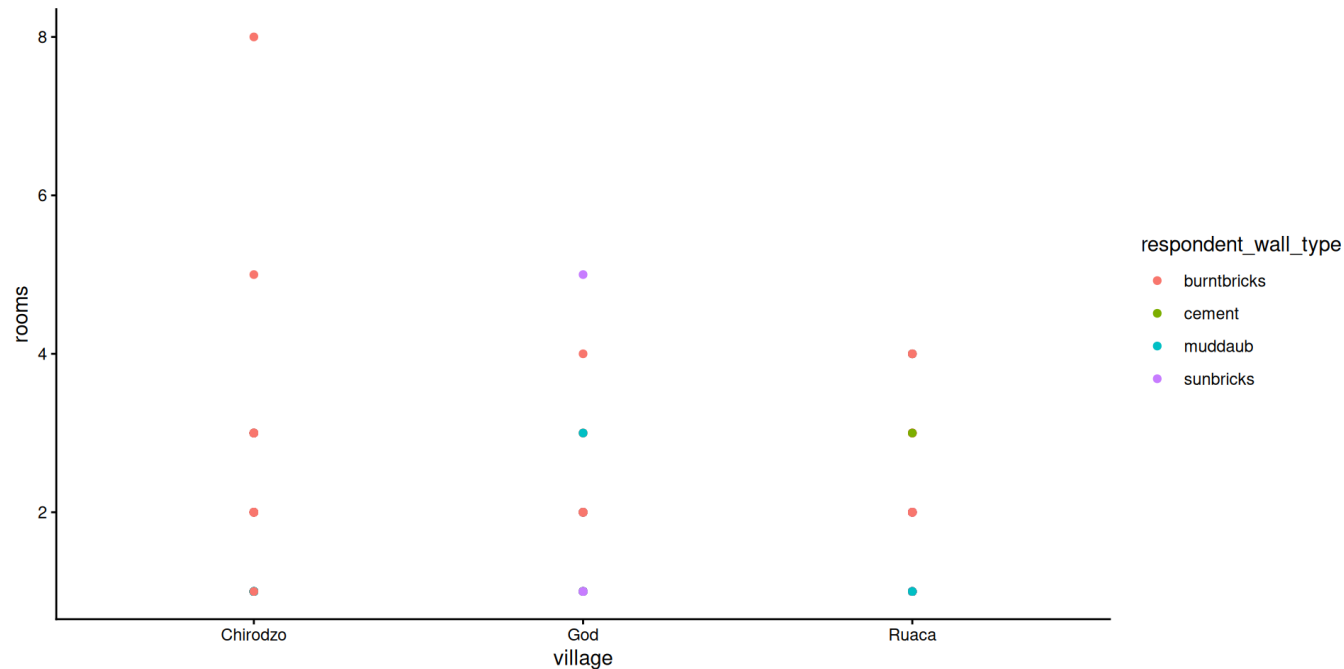


Figure 1: This plot shows the relationship between the variables room and village, but doesn't do a very good job at it.

Exercise 2

 4 mins

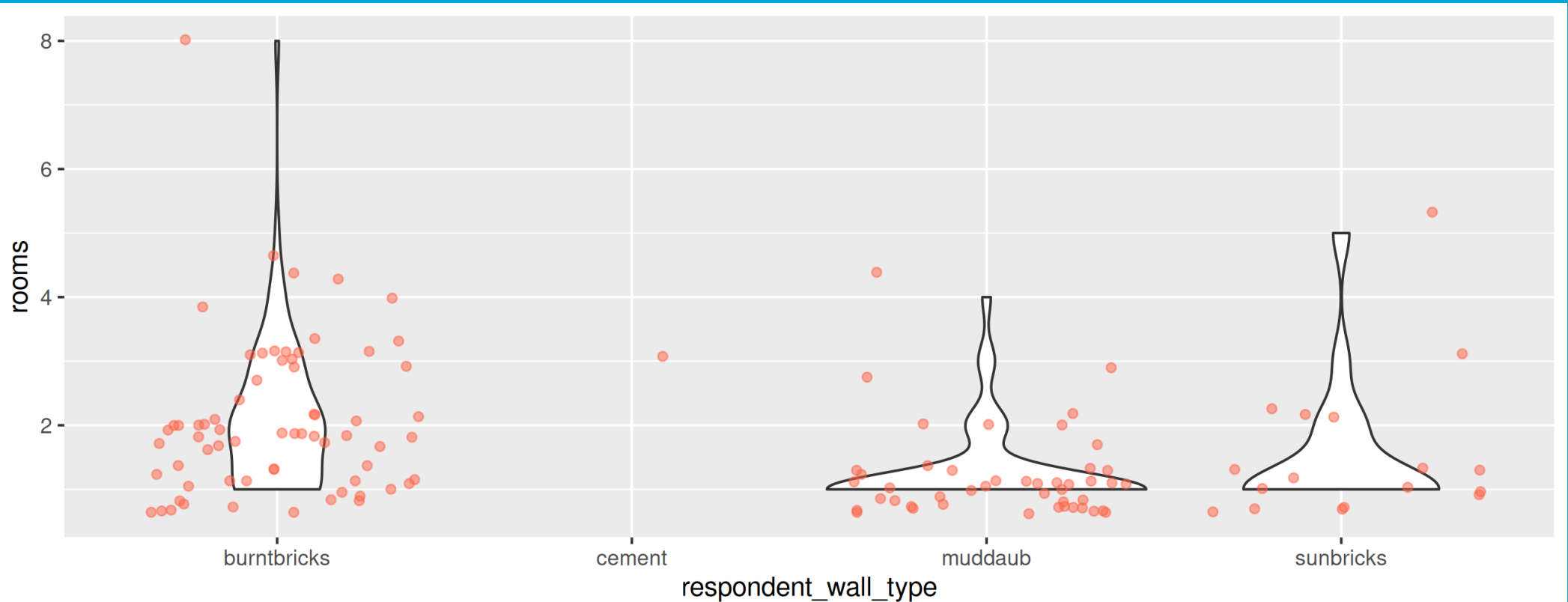
Boxplots are useful summaries, but hide the shape of the distribution. For example, if the distribution is bimodal, we would not see it in a boxplot.

Replace the box plot with a violin plot
see `geom_violin()`

04:00

Exercise 2: Solution

```
1 interviews_plotting %>%  
2   ggplot(aes(x = respondent_wall_type, y = rooms)) +  
3   geom_violin() +  
4   geom_jitter(alpha = 0.5, color = "tomato")
```



Exercise 3

 8 mins

Create a bar plot showing the proportion of respondents in each village who are or are not part of an irrigation association (`memb_assoc`).

Include only respondents who answered that question in the calculations and plot.

Which village had the lowest proportion of respondents in an irrigation association?

08:00

Exercise 3

 8 mins

Create a bar plot showing the proportion of respondents in each village who are or are not part of an irrigation association (`memb_assoc`).

Include only respondents who answered that question in the calculations and plot.

Which village had the lowest proportion of respondents in an irrigation association?

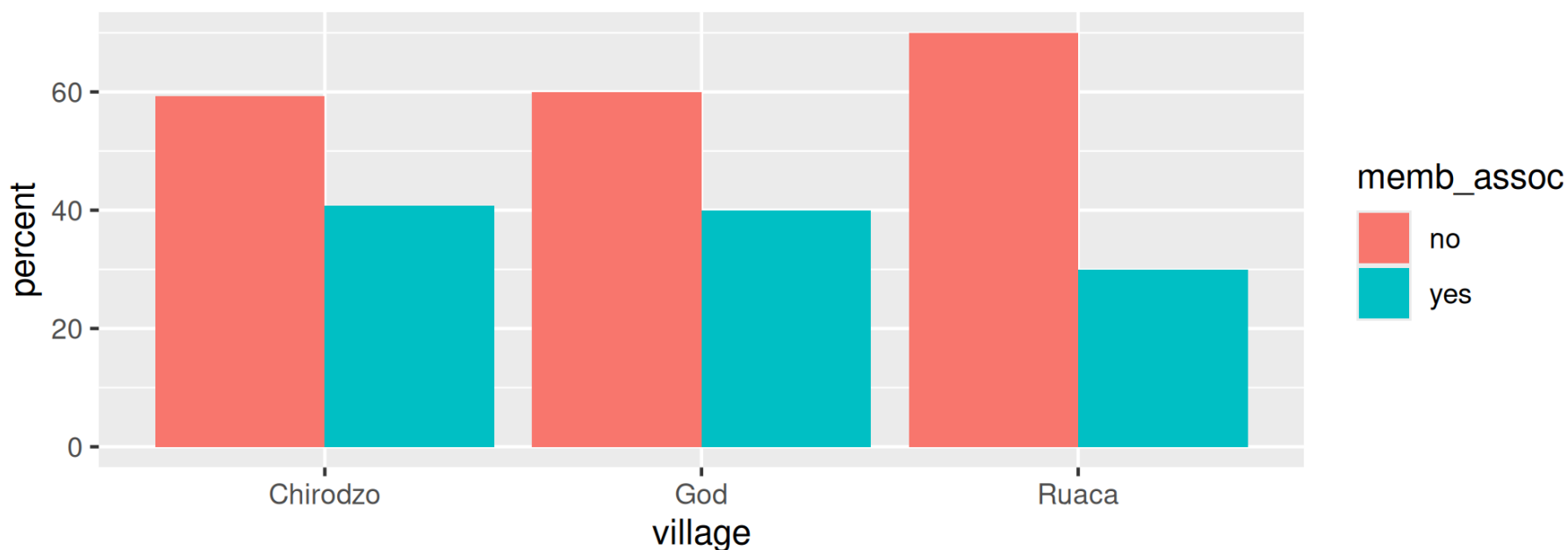
Hint

```
1 percent_memb_assoc <- interviews_plotting %>%  
2   filter(!is.na(memb_assoc)) %>%  
3   count(village, memb_assoc) %>%  
4   group_by(village) %>%  
5   mutate(percent = (n / sum(n)) * 100) %>%  
6   ungroup()
```

08:00

Exercise 3: Solution

```
1 percent_memb_assoc <- interviews_plotting %>%  
2   filter(!is.na(memb_assoc)) %>%  
3   count(village, memb_assoc) %>%  
4   group_by(village) %>%  
5   mutate(percent = (n / sum(n)) * 100) %>%  
6   ungroup()  
7  
8 percent_memb_assoc %>%  
9   ggplot(aes(x = village, y = percent, fill = memb_assoc)) +  
10  geom_bar(stat = "identity", position = "dodge")
```



Exercise 4

 4 mins

Experiment with at least two different themes. Build the previous plot using each of those themes.

Which do you like best?

Hint

```
theme_minimal  
theme_void  
theme_classic
```

```
theme_dark  
theme_grey  
theme_light
```

04:00

Exercise 4: Solution

```
1 percent_items %>%
2   ggplot(aes(x = village, y = percent)) +
3   geom_bar(stat = "identity", position = "dodge") +
4   facet_wrap(~ items) +
5   theme_bw() +
6   theme(panel.grid = element_blank())
```

